



# Digital Signal Processing

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Lecture 1

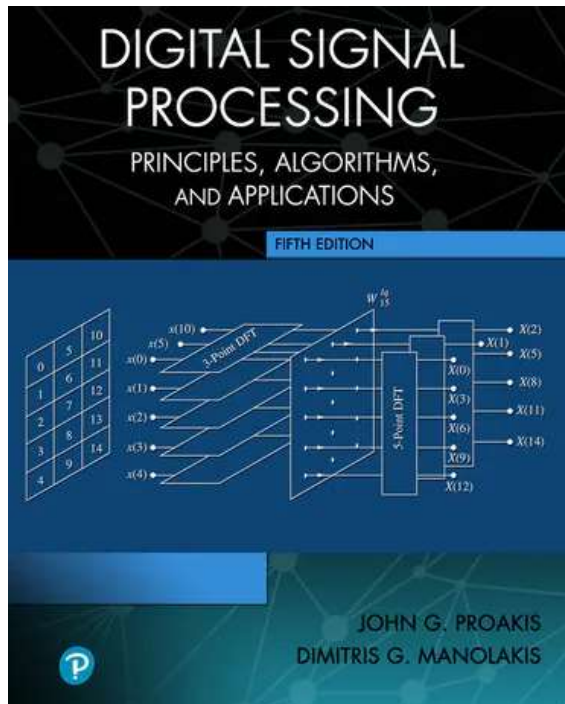
# Course Policy



# Course Outline

- Introduction to Digital Signal Processing.
- Sampling and Reconstruction.
- Discrete-Time Signals and Systems.
- Fourier analysis and its transform.
- Perform  $Z$  transform
- Design digital filters both FIR, IIR filters.
- DSP Algorithms and Implementation.

# References



Stein, Jonathan Y. "A computer science perspective". Wiley.

“Digital Signal Processing: Principles, Algorithms, and Applications” by John G. Proakis and Dimitris G. Manolakis. 5<sup>th</sup> Edition

# Evaluation

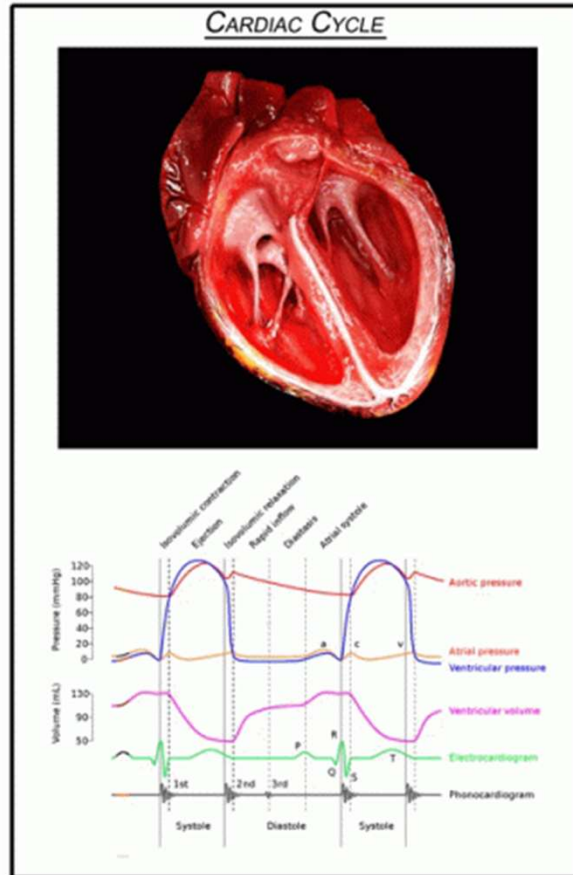
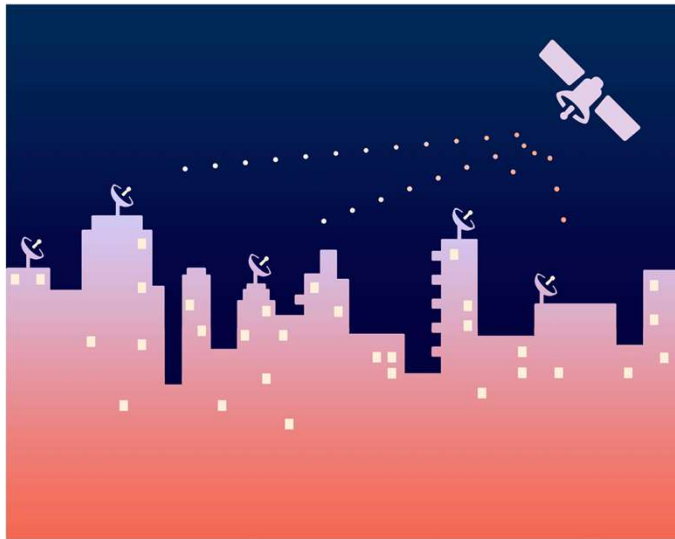
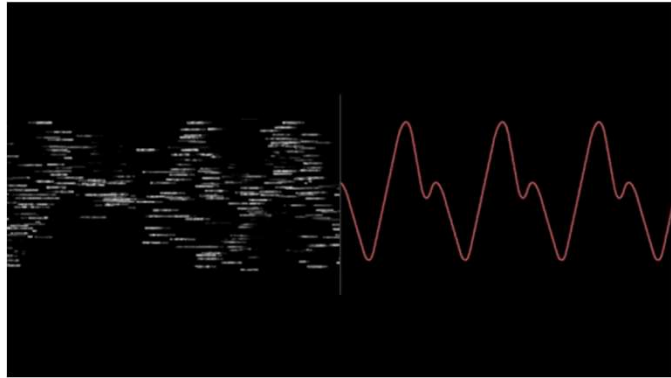
## Criteria

- Final Exam
- Practical
- Midterm
- Yearwork
  - Package
  - Quiz
  - Lecture Assessment

## Grade

- 50
- 20
- 15
- 15

# What is Signal?.



# Introduction

- **Signals**, like materials, are **real physical objects** but are **intangible, that** carry Information.
- Signals can be defined as physical quantities that **vary** in **time**, **medium**, and any **independent variable(s)**.
- **Not** all signals can be formed in **mathematical forms**

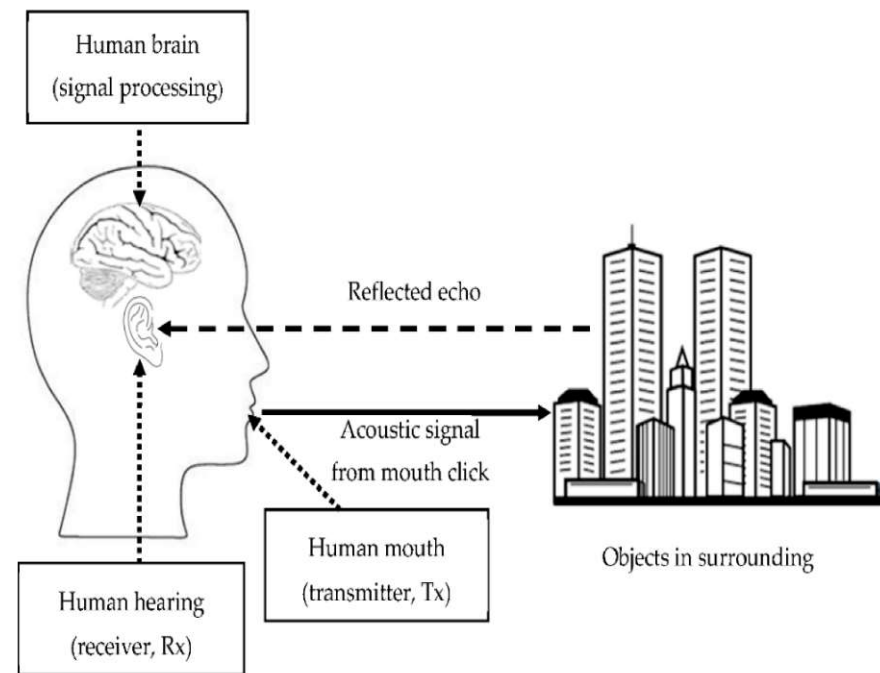
## Signal

Any physical quantity that varies as a function of time, space, or any other (independent) variable(s).

# Introduction

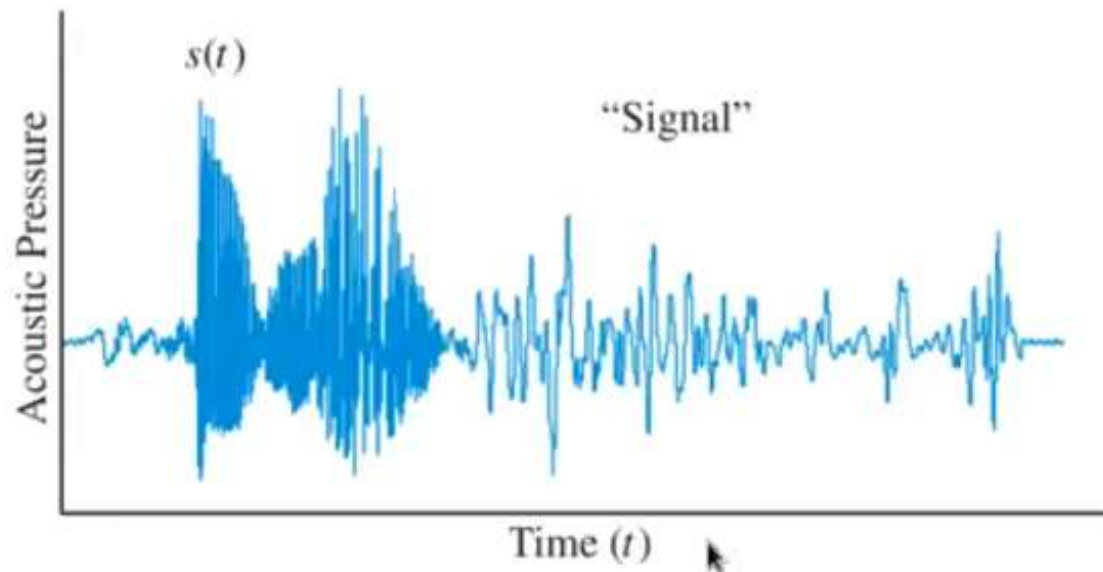
Humans are all remarkably **complex signal processing systems**, processing difficult audio and video signals every moment of our lives.

**Extracting** high-level **representations** of information carried by these **signals**, making decisions based on this information, storing/recording some of the information for later recall and processing, and **producing** new signals to change our environment in real time.



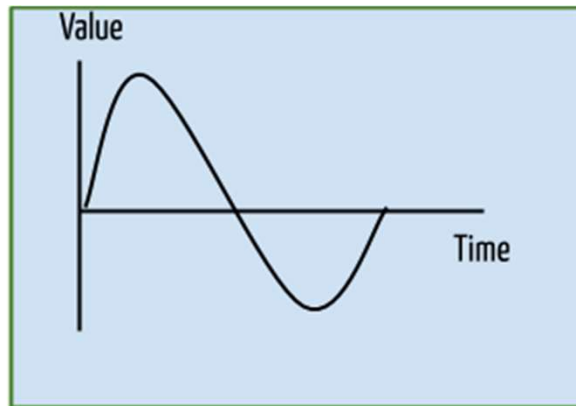
# Introduction Cont.

One example of signals is speech signals which represent variations of acoustic pressure, converted into electric signal by a microphone.

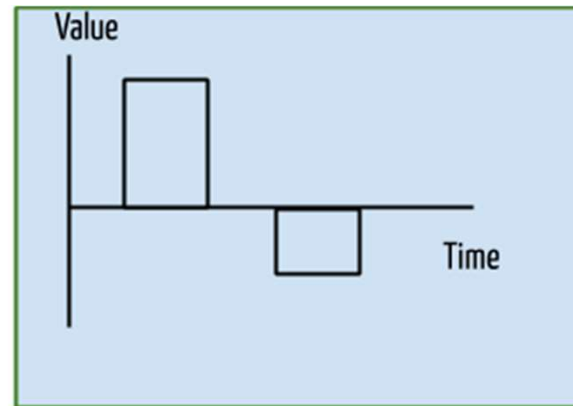


# What is Digital?

- Digital describes any system based on **discontinuous** (Discrete) data.
- Computers are digital machines because, at their most basic level, they can distinguish between just two values, 0 and 1, or off and on.



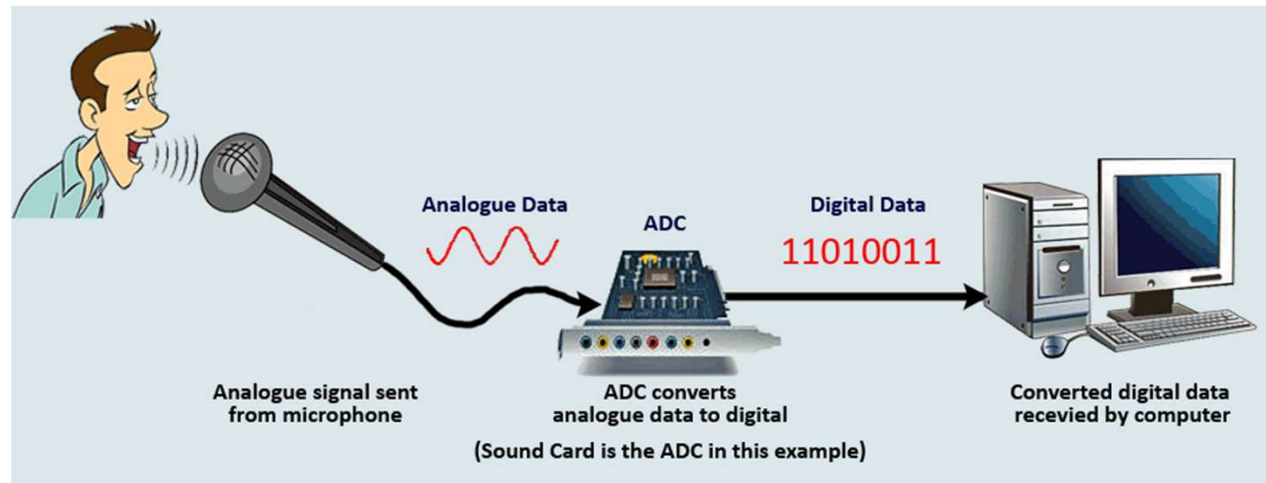
Analog Signal



Digital Signal

# Processing of Signals

- **Signal processing** is the task of **detecting, manipulating, and extracting** information from signals that have been **digitalized**.
- Signals are processed to display, analyze, or convert them into **another type** of signal for enabling the usage of **programming** instead of **analog electronic circuits**.



# Processing of Signals Cont.

- Signal could be presented in One or Multiple **Dimensions**.

## **One-Dimensional Signals:**

- **Audio Signals:** These are time-domain signals where the amplitude varies with time.
- **Temperature Recording:** A single sensor measuring temperature over time produces a one-dimensional signal.

## **Multi-Dimensional Signals:**

- **Images:** Digital images are two-dimensional signals where the intensity of each pixel is a function of two spatial variables (x and y coordinates).
- **Videos:** Videos are three-dimensional signals, with two spatial dimensions (x and y) and one temporal dimension (time).

# Processing of Signals Cont.

- Also, It could have one or more **Channels**.

## Single-Channel Signals:

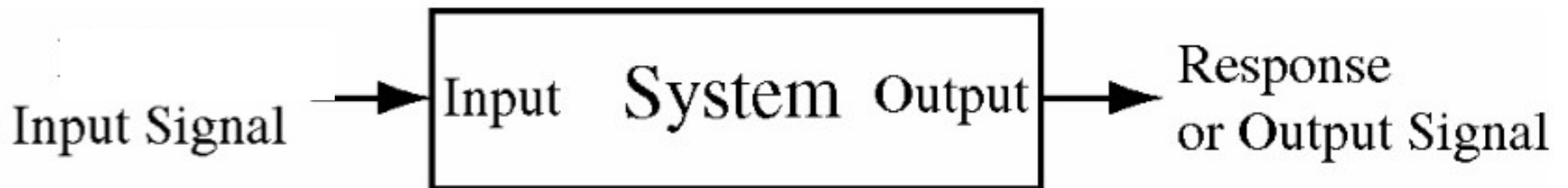
- **Room Temperature Recording:** A single sensor records the temperature of a room over time, producing a one-dimensional signal.
- **Mono Audio:** The audio output from a mono speaker.

## Multi-Channel Signals:

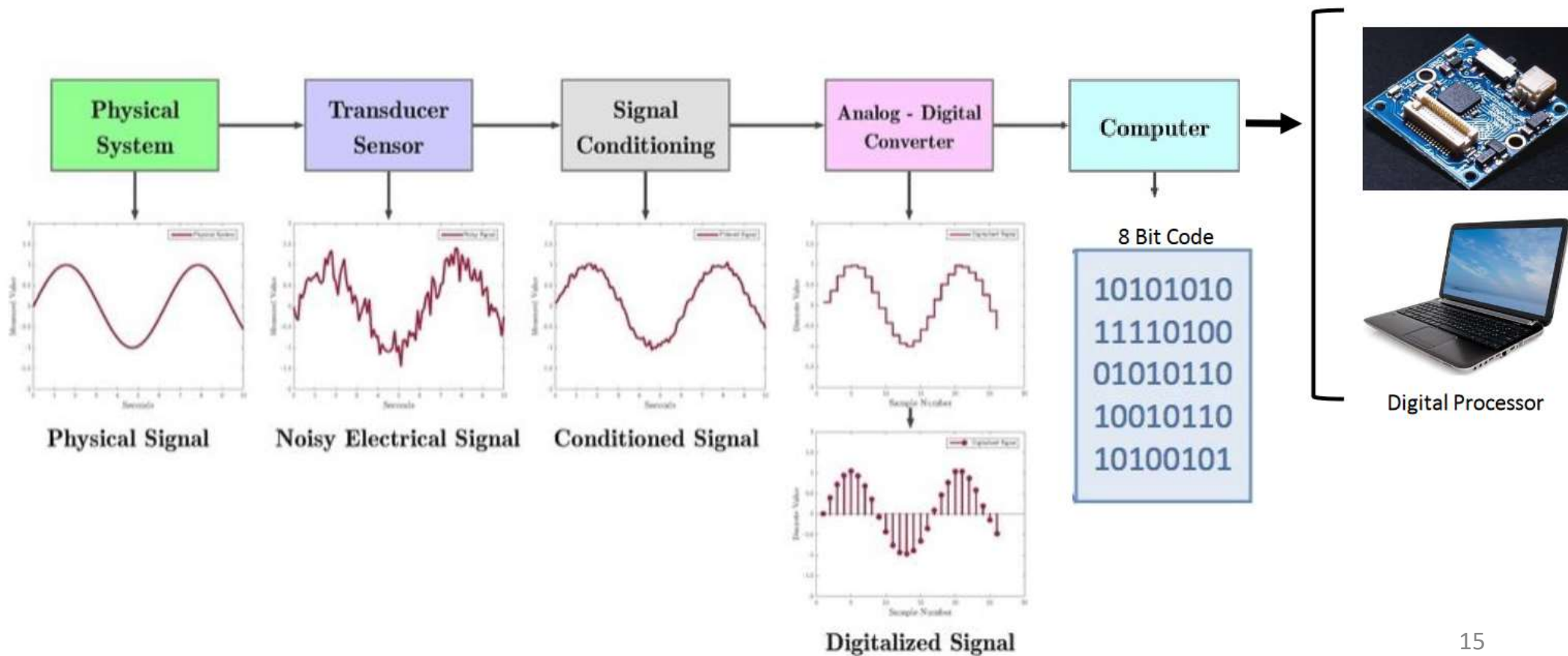
- **Electrocardiogram (ECG):** Often recorded using multiple leads (e.g., 3-channel or 12-channel ECG), each channel captures the electrical activity of the heart from different angles.
- **Stereo Audio:** In stereo sound systems, two channels (left and right) are used for enhancing the listening experience.
- **Digital images:** use three channels (red, green, and blue) to represent color information.
- **Electroencephalography (EEG):** uses multiple electrodes placed on the scalp to record electrical activity in the brain, with each electrode representing a different channel.

# Signal Processing System

- **System** is a device or combination of devices, which can operate on signals and produces corresponding response.
- **DSP System** is a software/hardware device that performs a function or operation on an input signal or signals.
- This can involve multiple mathematical operations such as compression, decompression, filtering, equalization, modulation, and demodulation to generate a signal of superior quality.



# Signal Processing System Flow



# Need to Remember

$$\pi \text{ radians} = 180^\circ$$

$$1 \text{ radian} = \frac{180^\circ}{\pi}$$

**Important**

Trigonometry Ratio Table

| Angles (In Degrees) | 0°          | 30°                  | 45°                  | 60°                  | 90°             | 180°        | 270°             | 360°        |
|---------------------|-------------|----------------------|----------------------|----------------------|-----------------|-------------|------------------|-------------|
| Angles (In Radians) | 0           | $\frac{\pi}{6}$      | $\frac{\pi}{4}$      | $\frac{\pi}{3}$      | $\frac{\pi}{2}$ | $\pi$       | $\frac{3\pi}{2}$ | $2\pi$      |
| sin                 | 0           | $\frac{1}{2}$        | $\frac{1}{\sqrt{2}}$ | $\frac{\sqrt{3}}{2}$ | 1               | 0           | -1               | 0           |
| cos                 | 1           | $\frac{\sqrt{3}}{2}$ | $\frac{1}{\sqrt{2}}$ | $\frac{1}{2}$        | 0               | -1          | 0                | 1           |
| tan                 | 0           | $\frac{1}{\sqrt{3}}$ | 1                    | $\sqrt{3}$           | Not Defined     | 0           | Not Defined      | 1           |
| cot                 | Not Defined | $\sqrt{3}$           | 1                    | $\frac{1}{\sqrt{3}}$ | 0               | Not Defined | 0                | Not Defined |
| csc                 | Not Defined | 2                    | $\sqrt{2}$           | $\frac{2}{\sqrt{3}}$ | 1               | Not Defined | -1               | Not Defined |
| sec                 | 1           | $\frac{2}{\sqrt{3}}$ | $\sqrt{2}$           | 2                    | Not Defined     | -1          | Not Defined      | 1           |

$$\sin(\pi - \theta) = \sin \theta$$

$$\cos(\pi - \theta) = -\cos \theta$$

$$\tan(\pi - \theta) = -\tan \theta$$

$$\sin(2\pi - \theta) = -\sin \theta$$

$$\cos(2\pi - \theta) = \cos \theta$$

$$\tan(2\pi - \theta) = -\tan \theta$$

$$\sin\left(\frac{3\pi}{2} - \theta\right) = -\cos \theta$$

$$\cos\left(\frac{3\pi}{2} - \theta\right) = -\sin \theta$$

$$\tan\left(\frac{3\pi}{2} - \theta\right) = \cot \theta$$

$$\sin^2 x + \cos^2 x = 1;$$

$$\sin^2 x = \frac{1 - \cos(2x)}{2};$$

$$\cos^2 x = \frac{1 + \cos(2x)}{2};$$

$$\sin(2x) = 2 \sin x \cos x;$$

$$\tan^2 x + 1 = \sec^2 x$$

# Need to Remember

$$\sin(-\theta) = -\sin \theta$$

$$\cos(-\theta) = +\cos \theta$$

$$\sin(\theta + 2\pi) = +\sin \theta$$

$$\cos(\theta + 2\pi) = +\cos \theta$$

$$e^{ix} = \cos x + i \sin x$$

$$e^{-ix} = \cos(-x) + i \sin(-x) = \cos x - i \sin x$$

$$\cos x = \operatorname{Re}\{e^{ix}\} = \frac{e^{ix} + e^{-ix}}{2}$$

$$\sin x = \operatorname{Im}\{e^{ix}\} = \frac{e^{ix} - e^{-ix}}{2i}$$



# Signal Basic Operations

Signal Basic Operations can be carried on

**Dependent Variables**

**Independent Variables**

# Signal Basic Operations

## For Dependent Variables (Amplitude)

- A. Amplitude scaling (Amplitude shifting, Amplification)
- B. Addition
- C. Subtraction
- D. Multiplication

## For Independent Variables (Time)

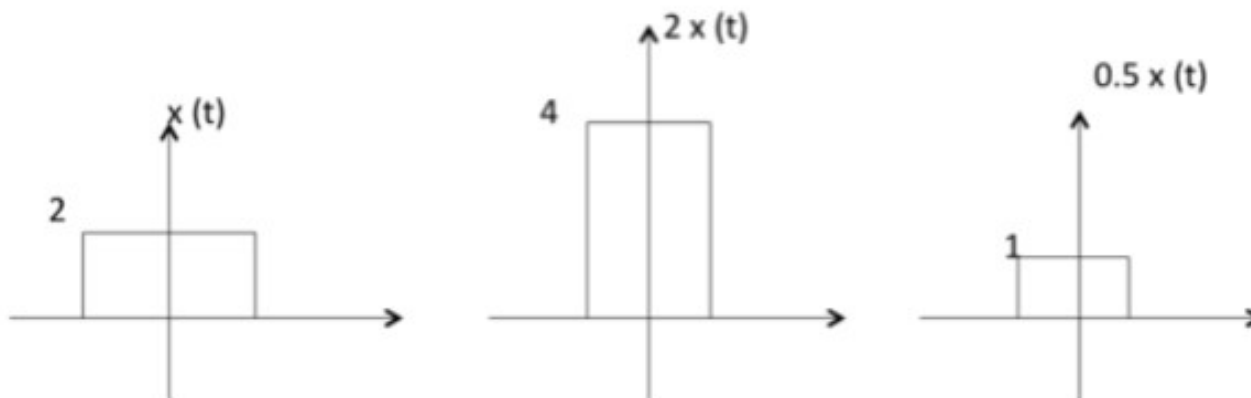
- A. Time Shifting
- B. Time Scaling
- C. Time Reversal (reflection, inversion, flipping)

# Signal Basic Operations

## For Dependent Variables (Amplitude)

### A. Amplitude scaling (Amplitude shifting, Amplification)

The scaled signal  $ax(t)$  is  $x(t)$  multiplied by the factor  $a$  where  $a$  is a constant real number, such as, the physical device that performs amplitude scaling is an electronic amplifier. Scaling can be up or down.



# Signal Basic Operations

## For Dependent Variables (Amplitude)

B. Addition: addition of two or more signals (their corresponding amplitudes).

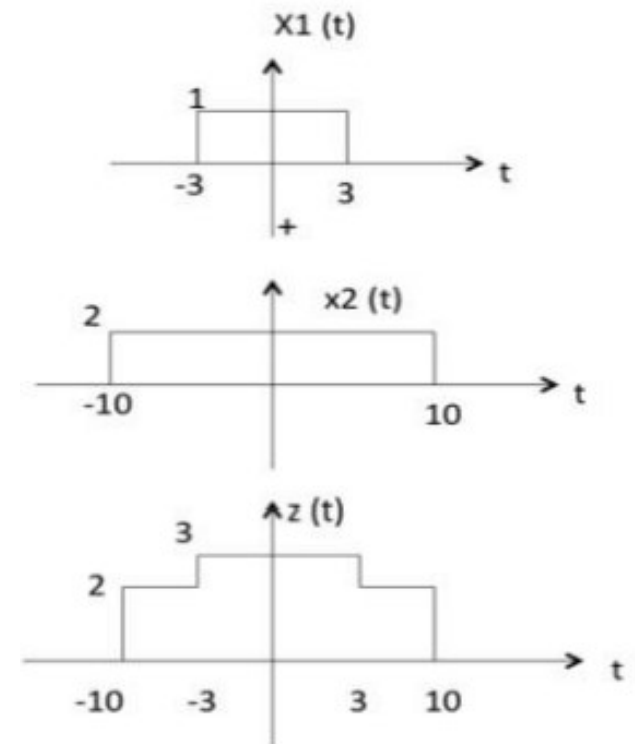
If  $x_1(t)$  and  $x_2(t)$  denote a pair of CTSs.

The signal  $z(t)$  obtained by the addition of  $x_1(t)$  and  $x_2(t)$  is defined by:

$$z(t) = x_1(t) + x_2(t)$$

In the case of DTS, it written as:

$$z[n] = x_1[n] + x_2[n]$$



# Signal Basic Operations

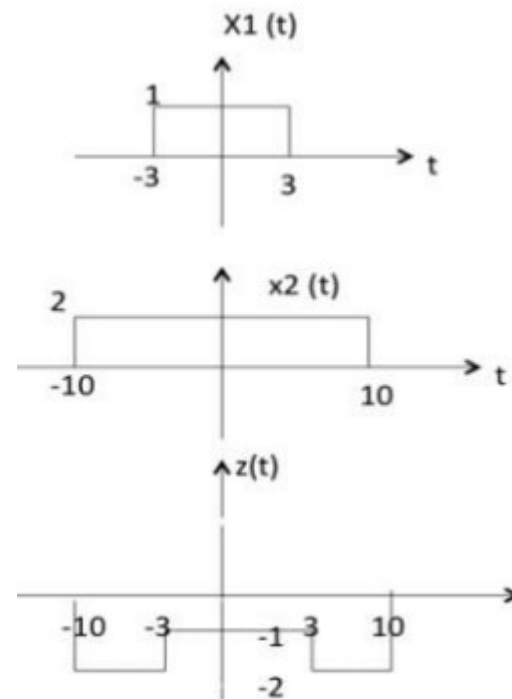
## For Dependent Variables (Amplitude)

- C. Subtraction: If  $x_1(t)$  and  $x_2(t)$  refer to a pair of CTSSs. Then, the signal  $z(t)$  obtained by the subtracting of  $x_1(t)$  from  $x_2(t)$  is defined by:

$$z(t) = x_1(t) - x_2(t)$$

In the case of DTS, it written as:

$$z[n] = x_1[n] - x_2[n]$$



# Signal Basic Operations

## For Dependent Variables (Amplitude)

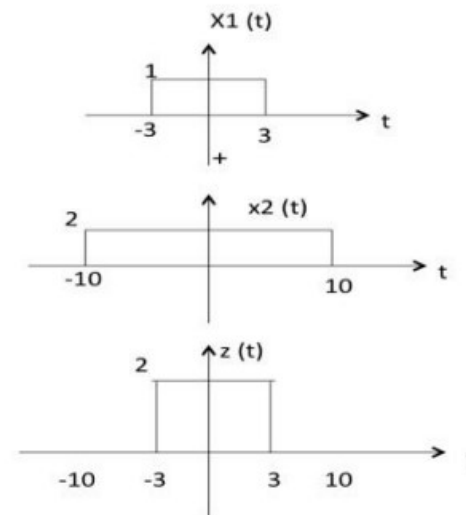
- D. Multiplication: let  $x_1(t)$  and  $x_2(t)$  denote a pair of CTSs. The signal  $z(t)$  resulting from the multiplication of  $x_1(t)$  and  $x_2(t)$  is defined by the following equation:

$$z(t) = x_1(t) * x_2(t)$$

That is, for each prescribed time ( $t$ ) the value of  $z(t)$  is given by the product of the corresponding values of  $x_1(t)$  and  $x_2(t)$ .

For discrete-time signals we write:

$$z[n] = x_1[n] x_2[n]$$



# Signal Basic Operations

## For Independent Variables (Time)

- A. Time Shifting: this kind of signal operation results in a positive or negative “shift” of the signal along its time axis. However, note that while doing so, none of its characteristics are altered.

If CTS is  $x(t)$ , then  $z(t)=x(t-T)$  is the signal  $x(t)$  shifted to the right by  $T$  units.

If CTS is  $x(t)$ , then  $z(t)=x(t+T)$  is the signal  $x(t)$  shifted to the left by  $T$  units.

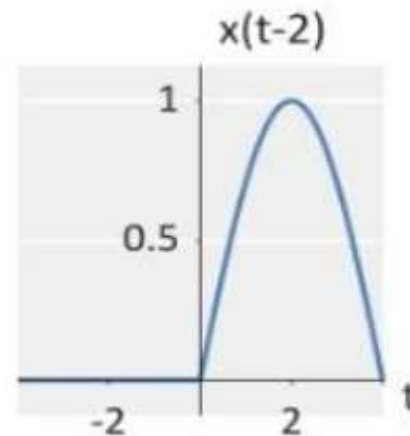
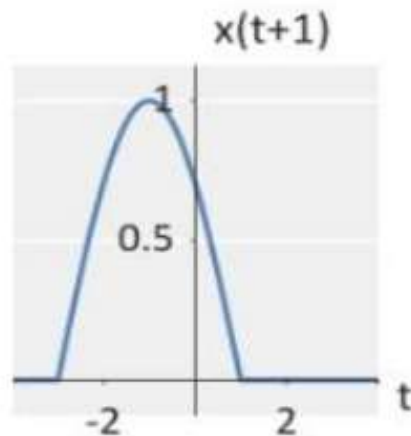
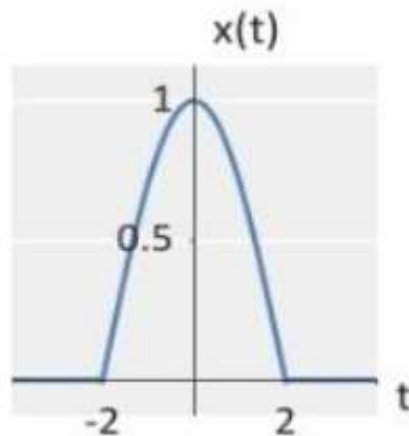
When DTS is  $x[n]$ , then  $z[n]=x[n-N]$  is the signal  $x[n]$  shifted to the right by  $N$  samples.

When DTS is  $x[n]$ , then  $z[n]=x[n+N]$  is the signal  $x[n]$  shifted to the left by  $N$  samples.

# Signal Basic Operations

## For Independent Variables (Time)

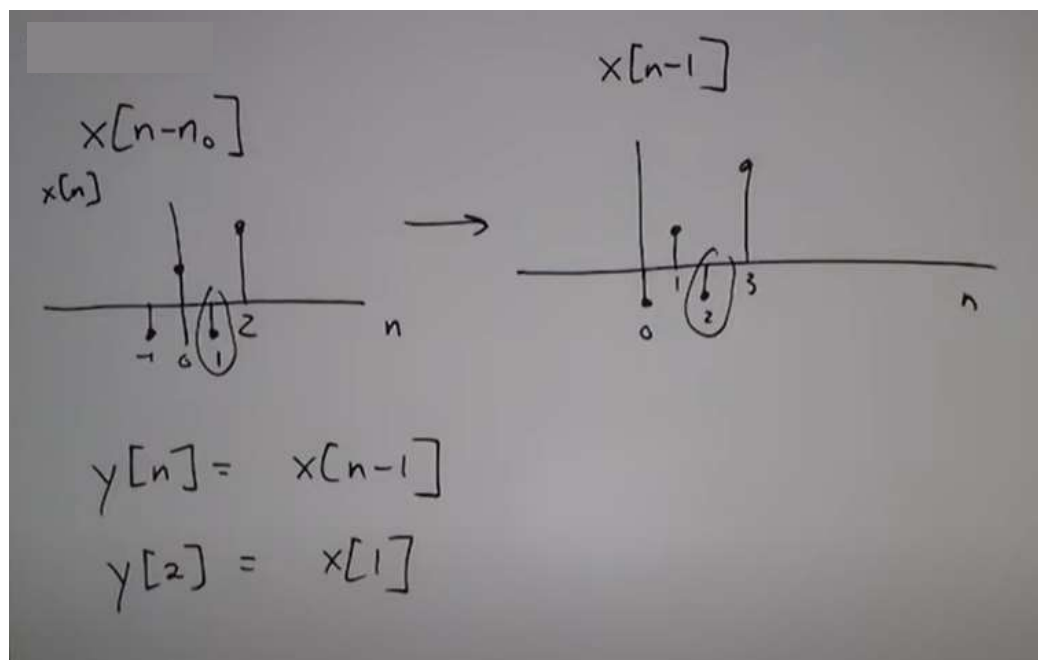
- A. Time Shifting: this kind of signal operation results in a positive or negative “shift” of the signal along its time axis. However, note that while doing so, none of its characteristics are altered.



# Signal Basic Operations

## For Independent Variables (Time)

- A. Time Shifting: this kind of signal operation results in a positive or negative “shift” of the signal along its time axis. However, note that while doing so, none of its characteristics are altered.



# Signal Basic Operations

## For Independent Variables (Time)

B. Time Scaling is a compression or expansion of a signal in time let  $x(t)$  denote a CTS, the signal  $z(t)$  obtained by scaling the independent variable, time ( $t$ ), by a factor “ $a$ ” is defined by two cases which are:

Case 1: If  $a > 1$ , the signal  $z(t)$  is a compressed version of  $x(t)$ . In this case:  $z(t) = x(at)$

Case 2: If  $a < 1$ , the signal  $z(t)$  is an expanded (stretched) version of  $x(t)$ . Thus, the resulted signal  $z(t)$  is computed as:  $z(t) = x(t/a)$

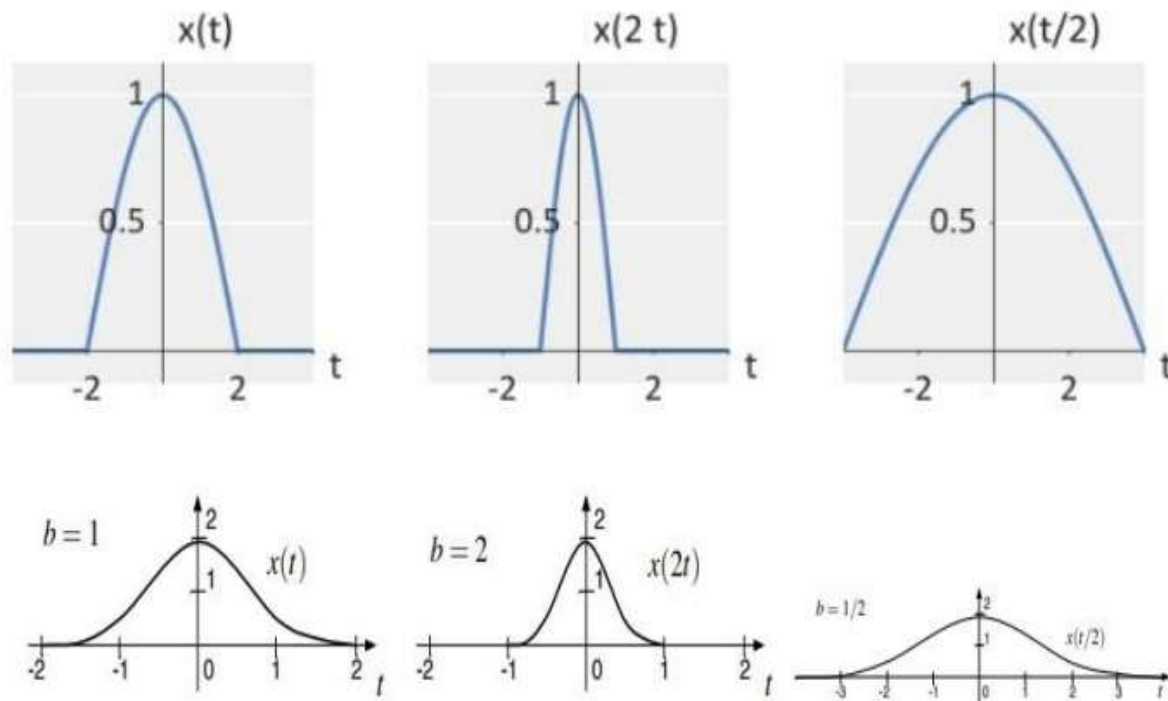
Note: the factor “ $a$ ” must not be equal to 0

# Signal Basic Operations

## For Independent Variables (Time)

B. Time Scaling is a compression or expansion of a signal in time.

All these cases are described in the below figure.



# Signal Basic Operations

## For Independent Variables (Time)

C. Time reversal (time inversion, reflection): The signal  $z(t)$  represents a reflected version of  $x(t)$  about the amplitude axis. Let  $x(t)$  denotes a CTS signal and  $z(t)$  denotes the signal obtained by replacing time  $(t)$  with  $(-t)$ , as shown in the next two cases:

- The CTS is represented as:

$$y(t)=x(-t)$$

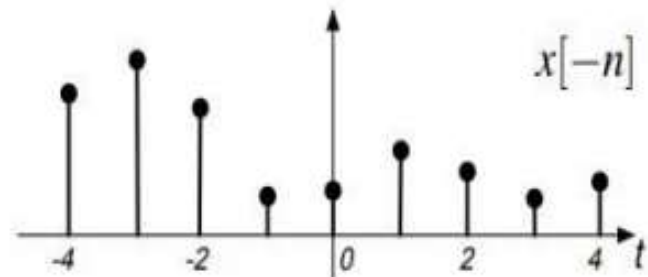
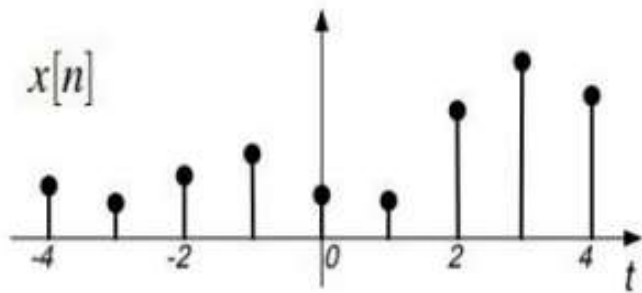
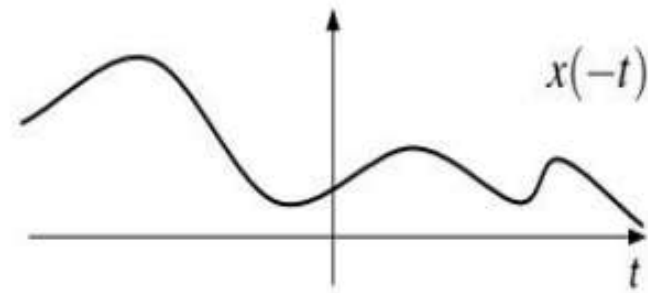
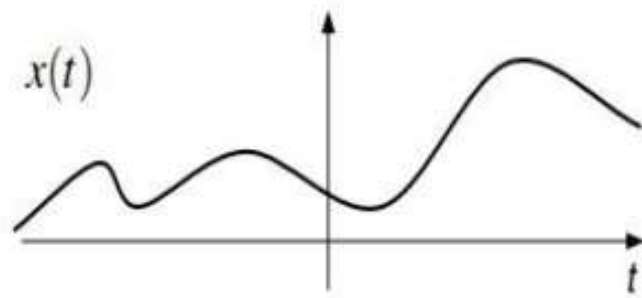
- While the DTS is as follows:

$$y[n]=x[-n]$$

# Signal Basic Operations

## For Independent Variables (Time)

### C. Time Reversal (reflection, inversion, flipping)



# Signals Classification

- 1. Analog vs. Digital Signals**
- 2. Continuous and Discrete Signals.**
- 3. Periodic and Aperiodic Signals (non-periodic).**
- 4. Even and Odd Signals.**
- 5. Deterministic and Random Signals.**
- 6. Causal, Anti-Causal and Non-Causal Signals.**
- 7. Others such as power, energy,...etc.**

# Signals Classification

Signals can be classified into several types based on different criteria. Here are some common classifications:

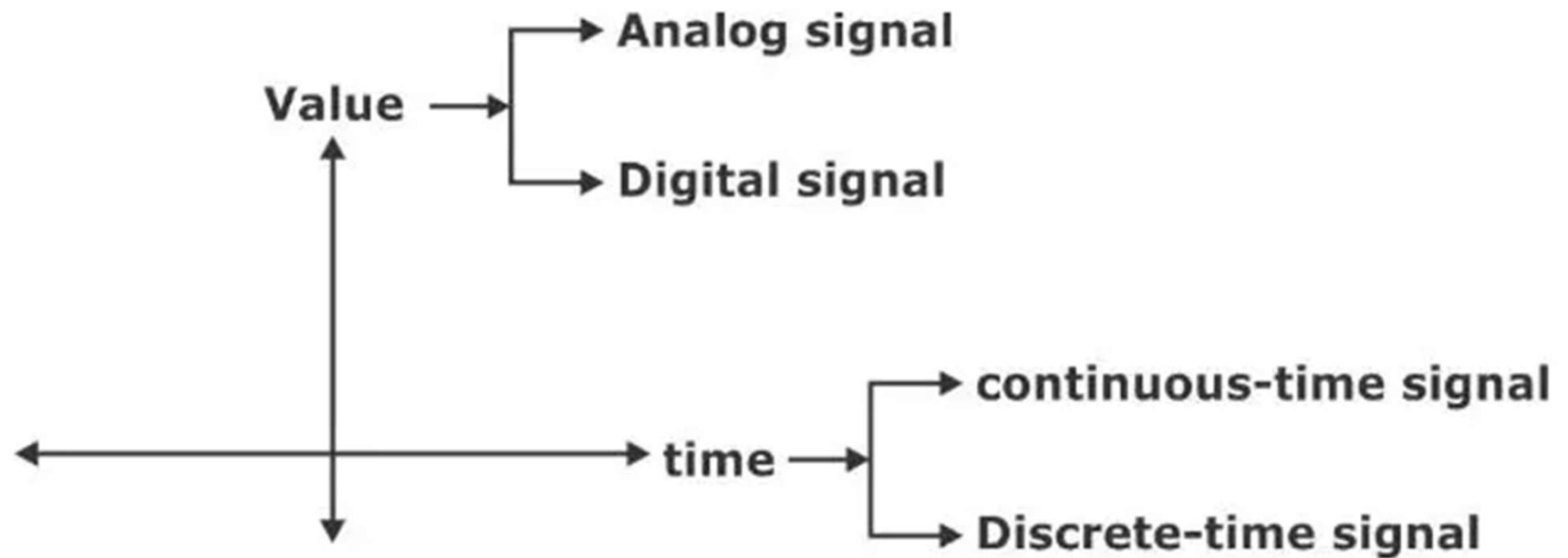
## 1. Based on Time

- **Continuous-Time Signals:** Defined for every instant of time. Example: Analog audio signals.
- **Discrete-Time Signals:** Defined only at discrete intervals of time. Example: Digital audio signals.

## 2. Based on Value

- **Analog Signals:** Have continuous values. Example: Temperature readings from a thermometer.
- **Digital Signals:** Have discrete values. Example: Binary data in computers.

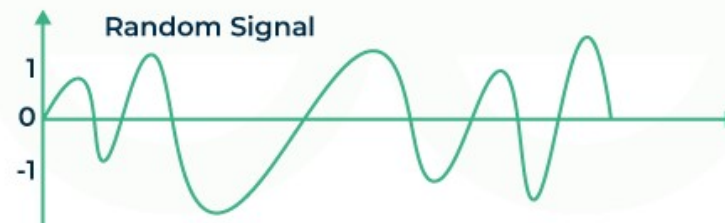
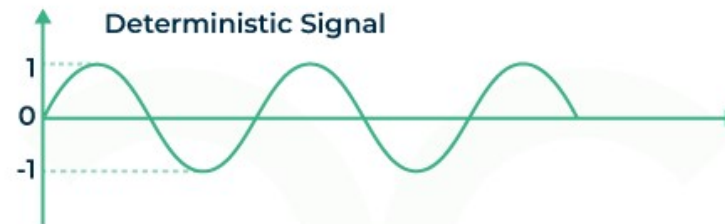
# Signals Classification



# Signals Classification

## 3. Based on Determinacy

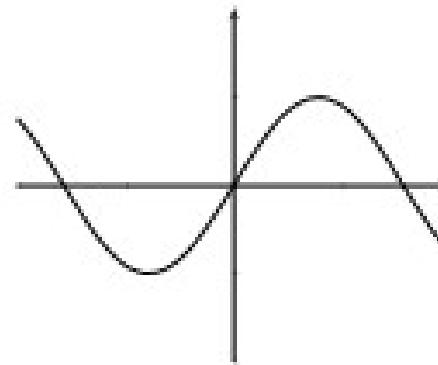
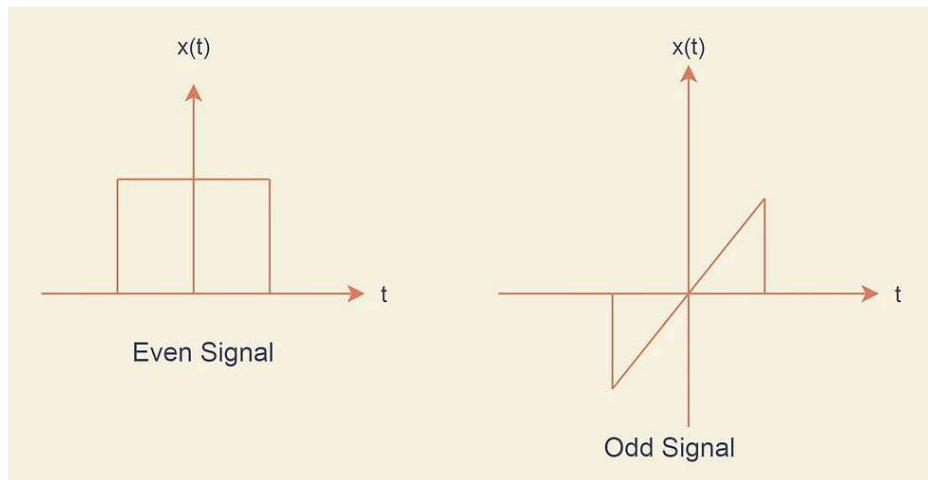
- **Deterministic Signals:** Can be precisely described by a mathematical expression. Example: A sine wave.
- **Random Signals:** Cannot be precisely described and are characterized by statistical properties. Example: Noise in communication systems.



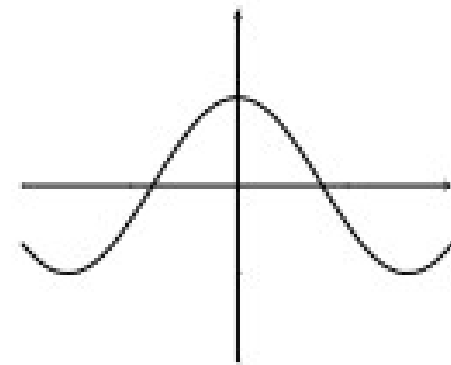
# Signals Classification

## 4. Based on Symmetry

- **Even Signals:** Symmetric around the vertical axis. Example: Cosine wave.
- **Odd Signals:** Antisymmetric around the vertical axis. Example: Sine wave.



$\sin x$  (odd)



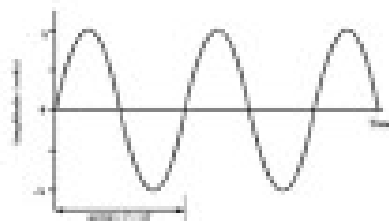
$\cos x$  (even)

# Signals Classification

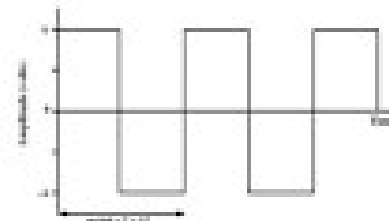
## 5. Based on Periodicity

- **Periodic Signals:** Repeat at regular intervals. Example: A square wave.
- **Aperiodic Signals:** Do not repeat at regular intervals. Example: A single pulse.

Periodic signals

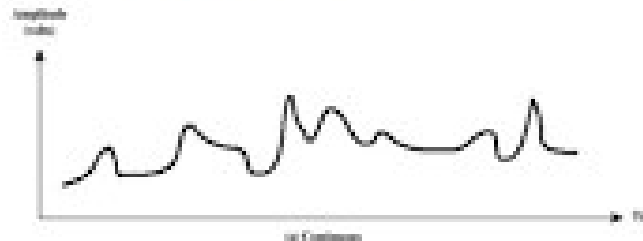


(a) Sine wave

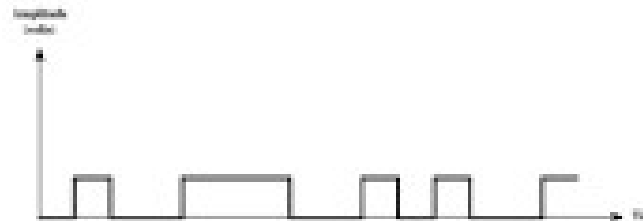


(b) Square wave

Nonperiodic signals



(a) Complex wave

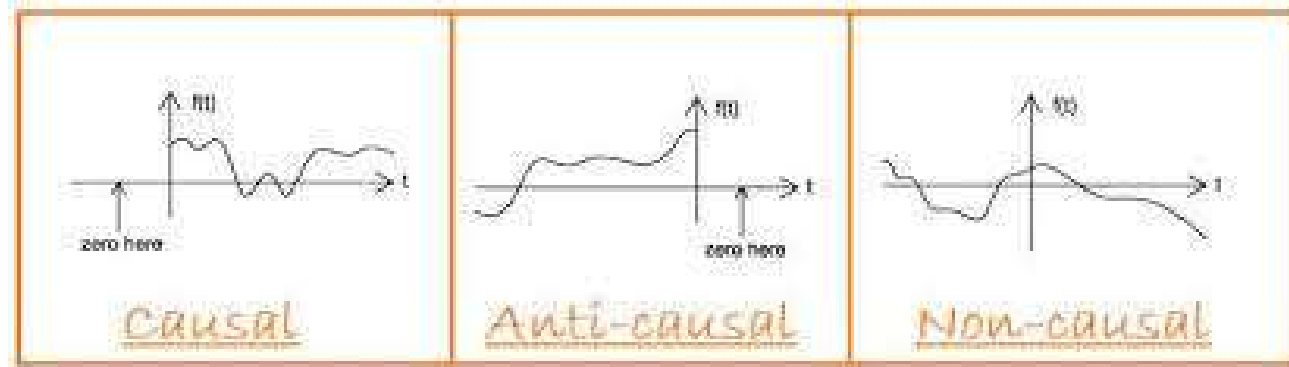


(b) Complex wave

# Signals Classification

## 6. Based on Causality

- **Causal Signals:** Zero for negative time. Example: Step function starting at ( $t = 0$ ).
- **Anti-causal Signals:** Zero for positive time. Example: Step function ending at ( $t = 0$ ).
- **Non-causal Signals:** Nonzero for both positive and negative times. Example: A sine wave.



# Continuous vs. Discrete vs. Digital Signals

## ANALOG SIGNAL

An analog signal is a continuous signal that changes over a time period.

A sine wave represents an analog signal.

## DIGITAL SIGNAL

A digital signal is a discrete signal that carries information in binary form.

A square wave represents a digital signal.

Fundamental frequency ( $f_0$ )

$$f_0 = \frac{1}{T_0}$$

Fundamental angular frequency ( $\omega_0$ )

$$\omega_0 = \frac{2\pi}{T_0} = 2\pi f_0$$

where (  $T$  ) is the period of the waveform.

Example: If a waveform repeats every 0.01 seconds, its fundamental frequency is:

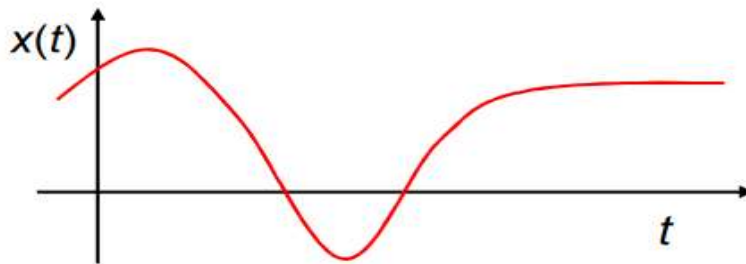
$$f_0 = 1/0.01 = 100\text{Hz} = 200 \pi \text{ Radian /second}$$

# Continuous vs. Discrete vs. Digital Signals

## Continuous-Time Signals

Most signals in the real world are continuous time, as the scale is infinitesimally fine.

Denote by  $x(t)$ , where the time interval may be bounded (finite) or infinite

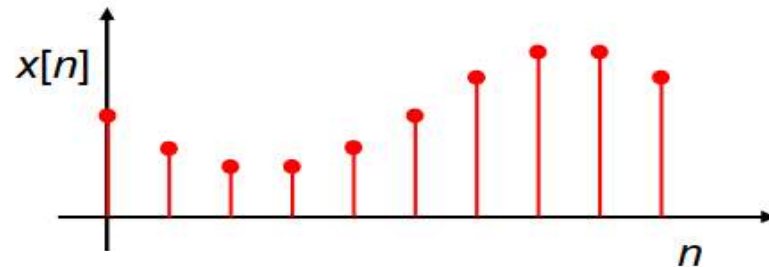


## Discrete-Time Signals

Some real world and many digital signals are discrete time, as they are sampled

E.g. pixels, daily stock price (anything that a digital computer processes)

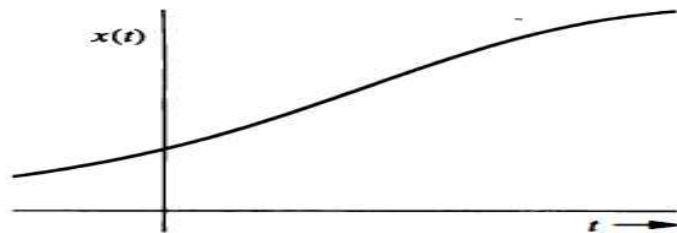
Denote by  $x[n]$ , where  $n$  is an integer value that varies discretely



## Sampled continuous signal

$x[n] = x(nT_s)$  where  $T_s$  is a sample time

# Continuous vs. Discrete vs. Digital Signals

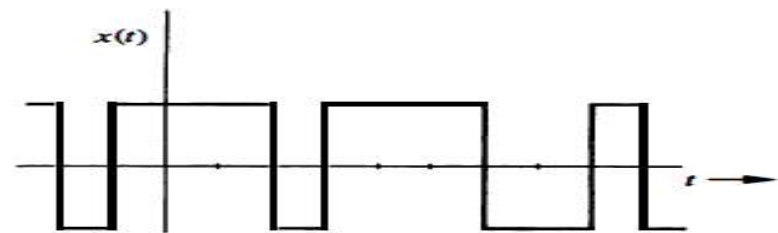


(a) analog, continuous time,

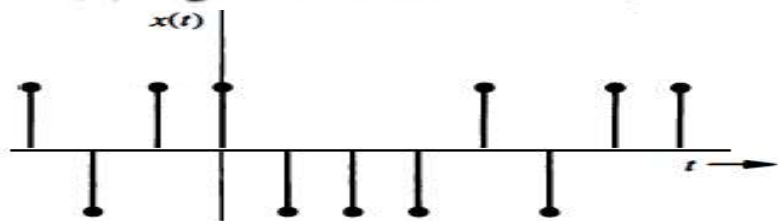


(c)

(c) analog, discrete time,



(b) digital, continuous time,



(d)

(d) digital, discrete time.

# Continuous vs. Discrete vs. Digital Signals

- Even though most of signals are in a continuous-time domain, they should be converted to a number at different discrete time to be processed by a microprocessor.
- The process of converting these signals into digital form is called analog-to-digital (A/D) conversion.

# Continuous vs. Discrete vs. Digital Signals

- The reverse process of reconstructing an **analog signal from its samples** is known as digital-to-analog (**D/A**) conversion.
- The most common techniques to change an analog signal to digital data include two steps: **Sampling and Quantization**.

# Continuous vs. Discrete vs. Digital Signals

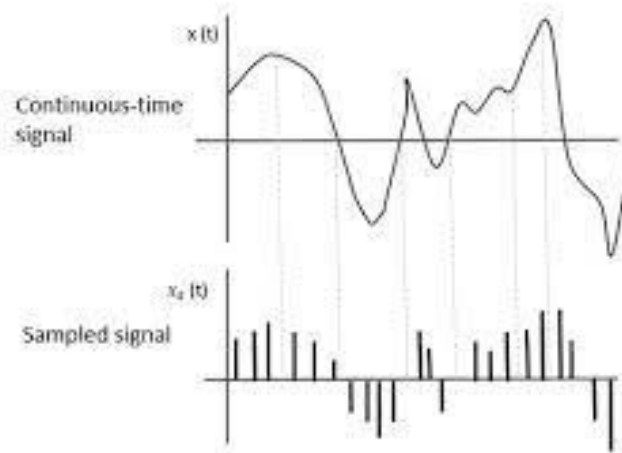
## So, what are digital signals?

- A digital signal is a discrete-time signal whose amplitude takes values from a **finite** set of real numbers.
- The “**Discreteness**” in amplitude is achieved through the quantization process.
- The quantization process results in the so-called quantization errors which can be reduced by increasing  $k$ .

# Sampling Process

**Sampling** is defined as, “The process of measuring the instantaneous **values of continuous-time (called samples)** signal in a discrete form.”

Sampling is the process in which continuous-time signals or analog signals are converted into discrete-time signals.



# Sampling Process

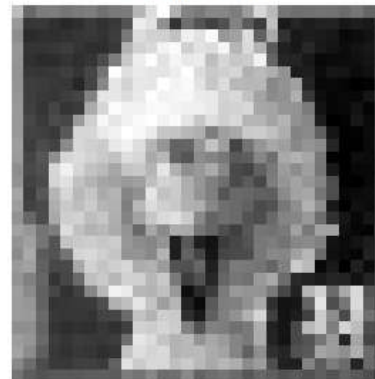
Sampling is associated with a term called **sampling rate**, which gives the number of samples per second, or sampling frequency.



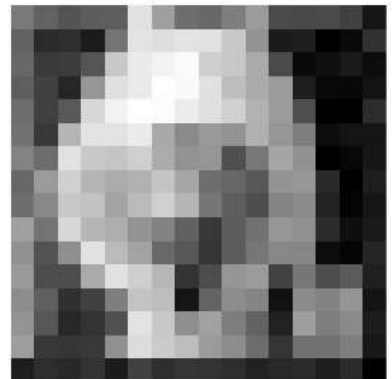
No subsampling



4 x 4 blocks



8 x 8 blocks



16 x 16 blocks

Sampling an image.

# Quantization

- During data storage or data transmission, we use **bit sequences** to represent real numbers.
- Since there are an infinite number of real numbers, it is not possible to represent this vast amount of real numbers by **limited-length bit** streams.
- For this reason, we choose a number of real numbers to represent by bit streams and try to round other real numbers to the chosen ones when it is necessary to represent them by bit streams.

# Quantization

Converting the continuous real values of the analog signal into bounded digital ones require rounding them to the nearest allowed level.

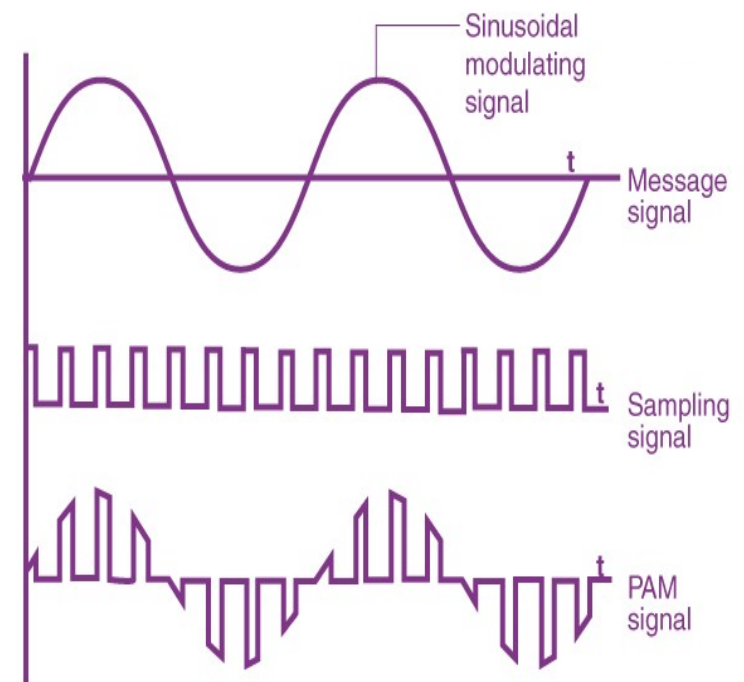
This will inevitably lead to a loss of precision, which can be interpreted as adding (real-valued) noise to each value

$$a = d_n + \Delta_{error}$$

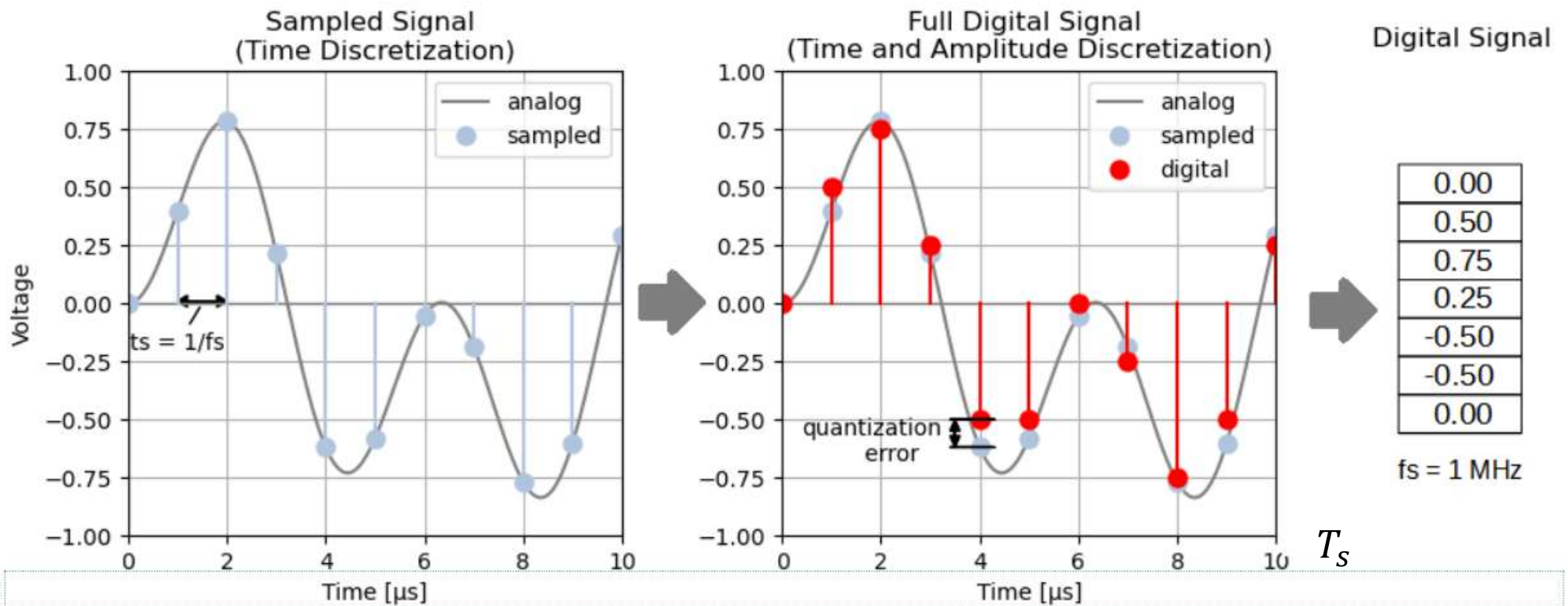
where  $\Delta_{error}$  can **never exceed half** the distance to nearby quantization levels.

# Quantization

- The effect of this noise is to degrade the Signal to Noise Ratio (SNR) of the signal.
- A degradation decreases in magnitude when the number of available levels is increased.
- Digital signals obtained from analog ones are sometimes called **Pulse Code Modulation (PCM)** streams.



# Sampling and Quantization



AD Conversion Process: Sampling with  $f_s = 1 \text{ MHz}$  and 3-bit quantization

# Sampling and Quantization

Periodic sampling, i.e.  $s[n] = s(t)|_{t=nT_s}$ , where  $T_s$  is the sampling period.

$$s[n] = s(t)|_{t=nT_s} = s(0), s(T_s), s(2T_s), \dots$$

Nyquist Condition

$$F_s > F_{max}$$

How to choose  $F_s = 1/T_s$ ?

The values of the discrete-time signal between two sampling instants is NOT zero, it is not defined.

( ): continuous independent variable (i.e. real numbers)

[ ]: discrete independent variable (i.e. integers only)

$s[n]$  : discrete-time signal  $\Leftrightarrow$  sequence of real numbers

$n$  is an integer



Any Question?  
Thank you